# Abstract

Class imbalance makes credit assessment and thyroid screening difficult to evaluate using accuracy alone. We compared decision trees, random forests, logistic regression and support vector classifiers on two UCI datasets. Each model was tested with plain training, SMOTE, class weighting and threshold adjustment. We used the same validation splits and fitted preprocessing within training folds. Apriori supplied association rules to help explain patterns in German Credit. Cost sensitive logistic regression achieved the highest mean F1 in the German Credit main experiment, while random forest with SMOTE led on sick. Strategies that improved recall could reduce precision, showing why accuracy alone can hide poor minority class performance. The additional ensemble study achieved a higher exploratory F1 on sick, but we did not directly test its advantage over the main experiment winner. Apriori found combinations associated with higher bad proportions and added to the tree model explanations. These results show that model choice depends on the dataset and decision objective. Independent validation is needed before practical use.


## 6.6 Association Rule Evidence

**Association rule evidence.** Apriori found simple combinations associated with the bad label and added to the supervised model explanations. The leading rule combined the highest duration band with savings below 100. Among 124 matching records, 71 were labelled bad. Support was 0.071, confidence 0.573 and lift 1.91, compared with an overall bad proportion of 0.30. We retained six rules with confidence of at least 0.50 and a gain of at least five percentage points over the stronger single condition. We discretised three continuous fields, then mined frequent itemsets from 1,000 Boolean transactions using minimum support 0.05 and at most three items per itemset. Some fields also appeared in decision tree paths and ranked highly in random forest importance, but the tree reproduced no complete selected rule. This overlap helps interpretation, but does not independently validate the rules. The rules were selected using all records and establish neither causation nor predictive accuracy.

## 7.2 Limitations

These results describe internal comparisons on two datasets using one split seed. They do not establish performance on new populations. Retained duplicate sick records may appear in different validation folds. Configuration selection used cross validation over all records, so the overlapping holdout does not provide an independent test of that selection. Ordinary SMOTE can create artificial categorical representations, but we did not isolate their effect on performance. The ensemble extension and Apriori analysis are exploratory. We did not directly test the higher ensemble F1 against the main experiment winner. These limits leave uncertainty about generalisation and the reasons for model differences.

## 7.3 Future Work

The next step is to test the selected configurations on independent data, or use nested cross validation with several seeds. A controlled comparison with SMOTENC could test whether preserving categorical values improves resampling. Threshold selection should follow the decision objective, including the official credit cost matrix where applicable. The Apriori rules also need checks on separate samples to see whether their support, confidence and lift remain stable.